Pointer to array:

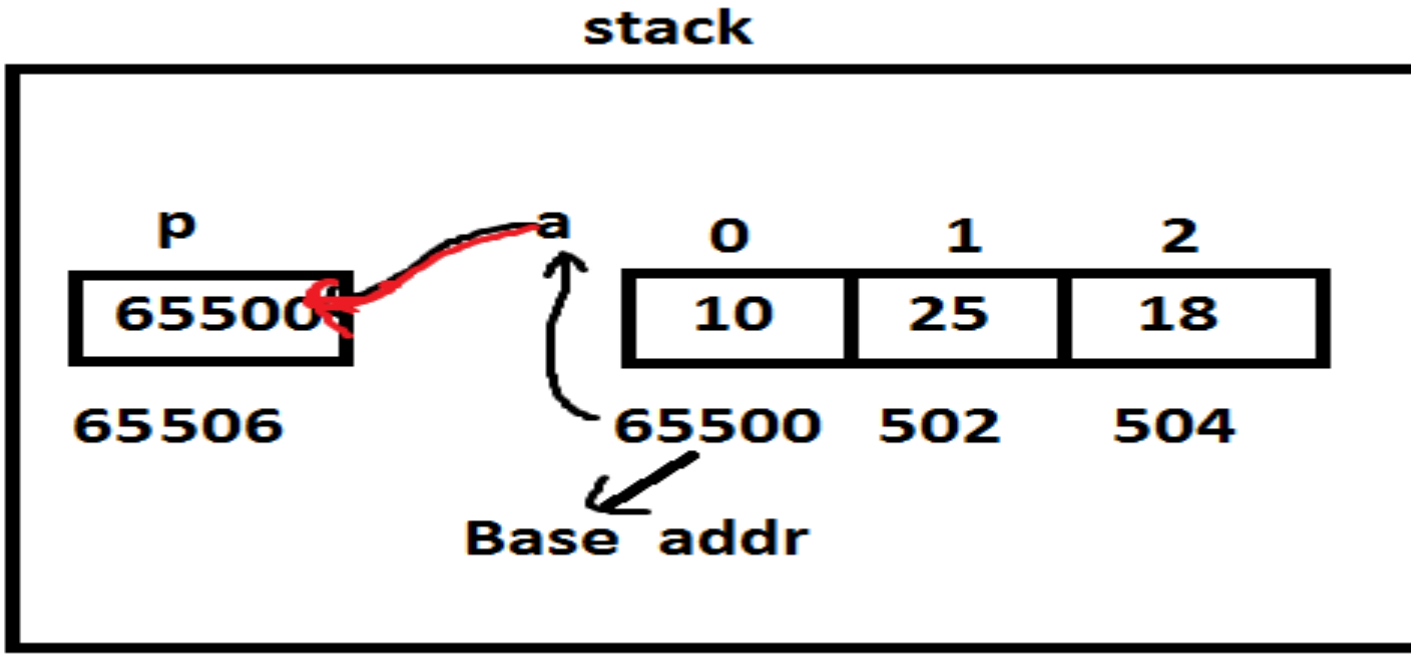
Array is implicit pointer. Due to this it holds the base cell addr [0 cell addr] implicitly. By assigning the array name or 0 cell addr to the pointer, we can handle array elements using the following syntax.

***(ptrvariable + offset/index * sizeof(variable));**

Eg:

int a[3]={10, 25, 18}, *p, i;

p = a ; or p = &a[0]; or p = &a;




```
for(i=0;i<3;i++)
printf("%4d", *(p+i));
```

Here *(p+i) meaning is:

p is 65500

1. *(p+0*2)➔*65500➔value at 65500➔**10**
2. *(p+1*2)➔*65502➔value at 65502➔**25**

3. *(p+2*2)➔*65504➔value at 65504➔18

Note: Here 2 is int size.

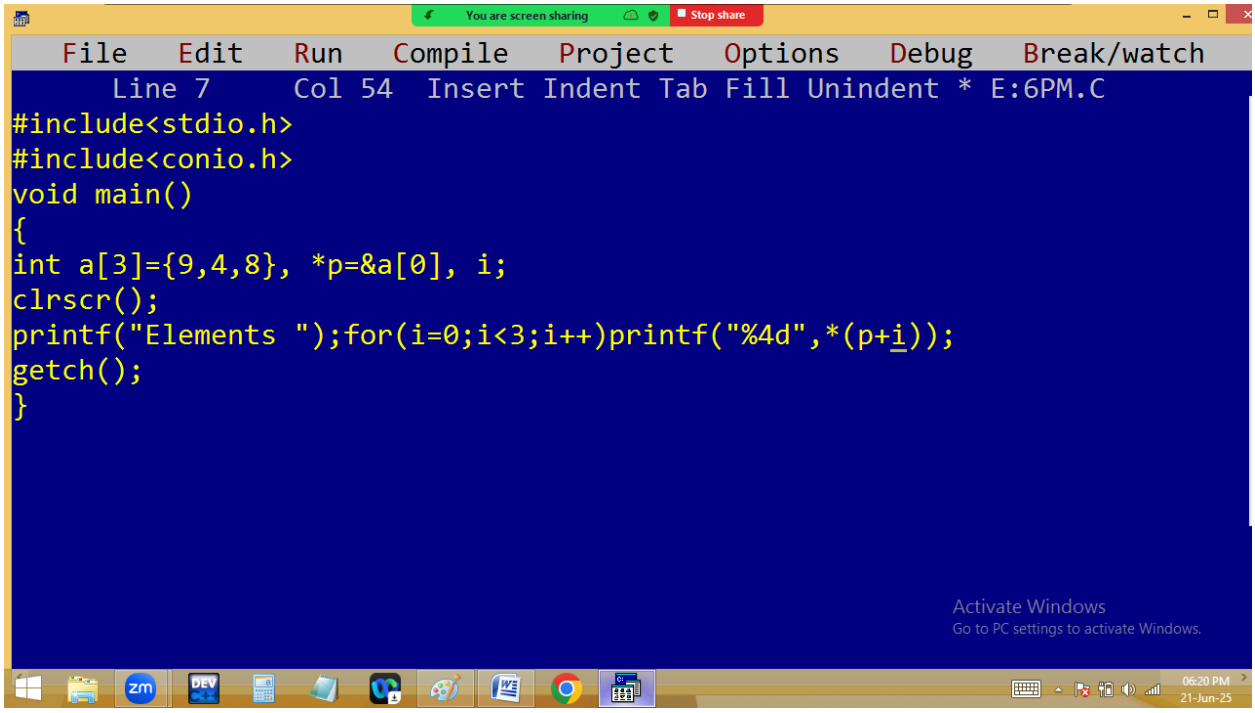
Eg:

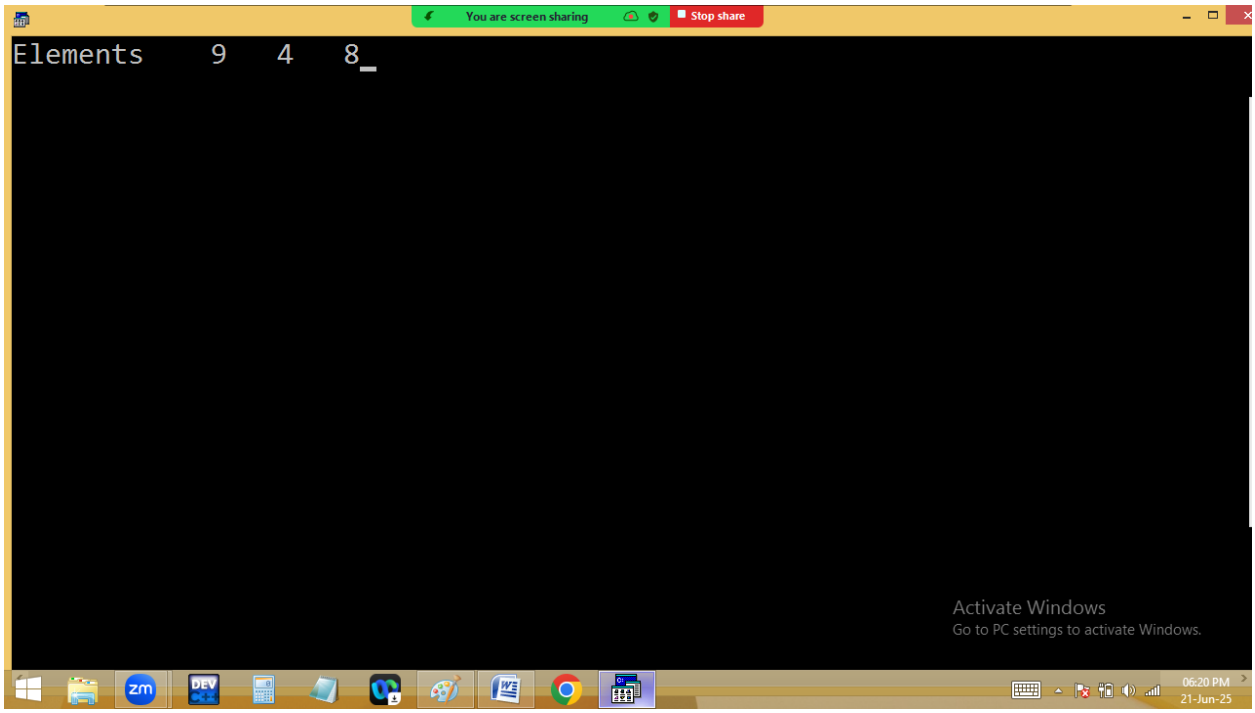
```
#include<stdio.h>
#include<conio.h>
void  main()
{
int  *p,  a[3]={10,25,18}, i;
clrscr();
p = a;   /* p=&a[0]; or p=&a; */
printf("Elements are: ");
for(i=0;i<3;i++)
printf("%4d",*(p+i));
getch();
}
```

Output: Elements are: 10 25 18

Note: We can access array elements using array / pointer in following ways.

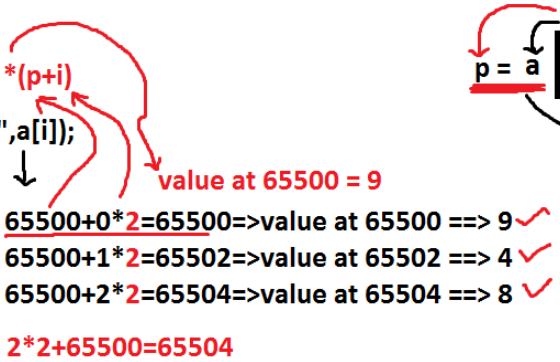
a[i] / i[a] / p[i] / i[p] / *(p+i) / *(a+i) / *(i+p) / *(i+a)

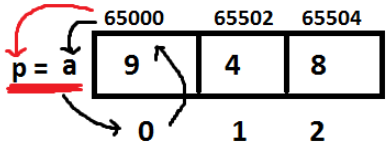

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File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 54 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[3]={9,4,8}, *p=&a[0], i;
clrscr();
printf("Elements ");for(i=0;i<3;i++)printf("%4d",*(p+i));
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:20 PM
21-Jun-25


You are screen sharing
Stop share
Elements 9 4 8
Activate Windows
Go to PC settings to activate Windows.
06:20 PM
21-Jun-25

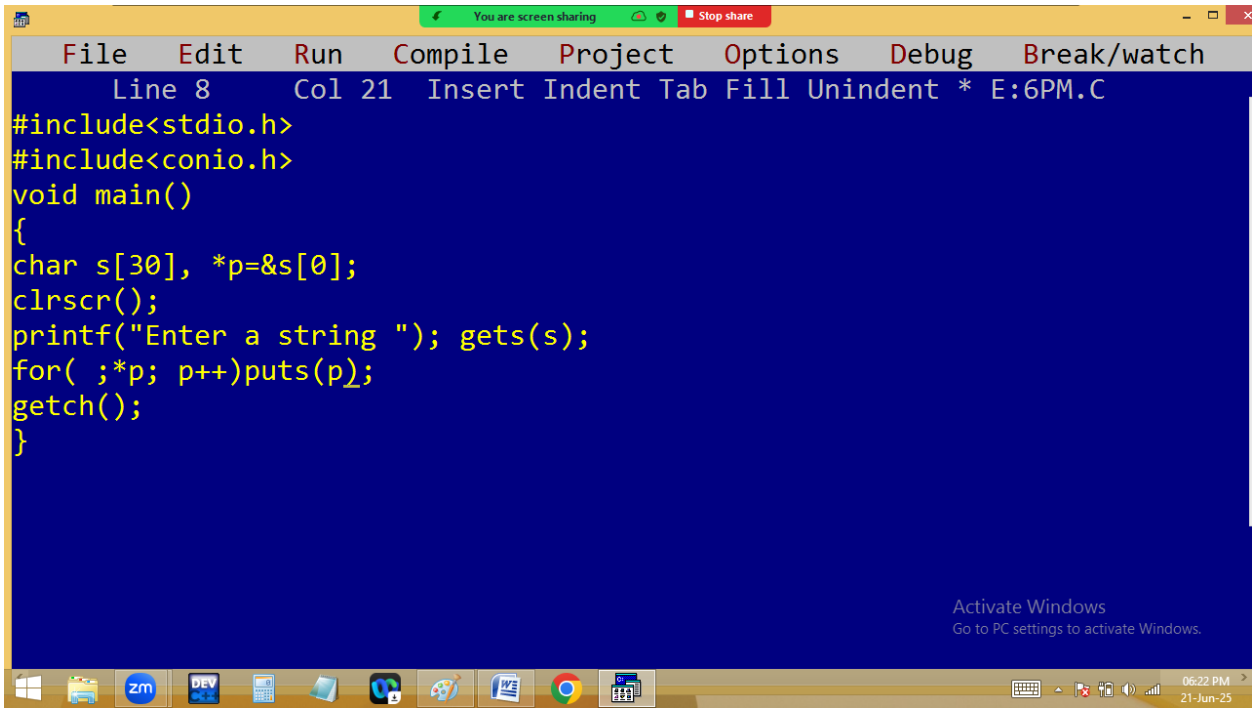
p = &a[0];

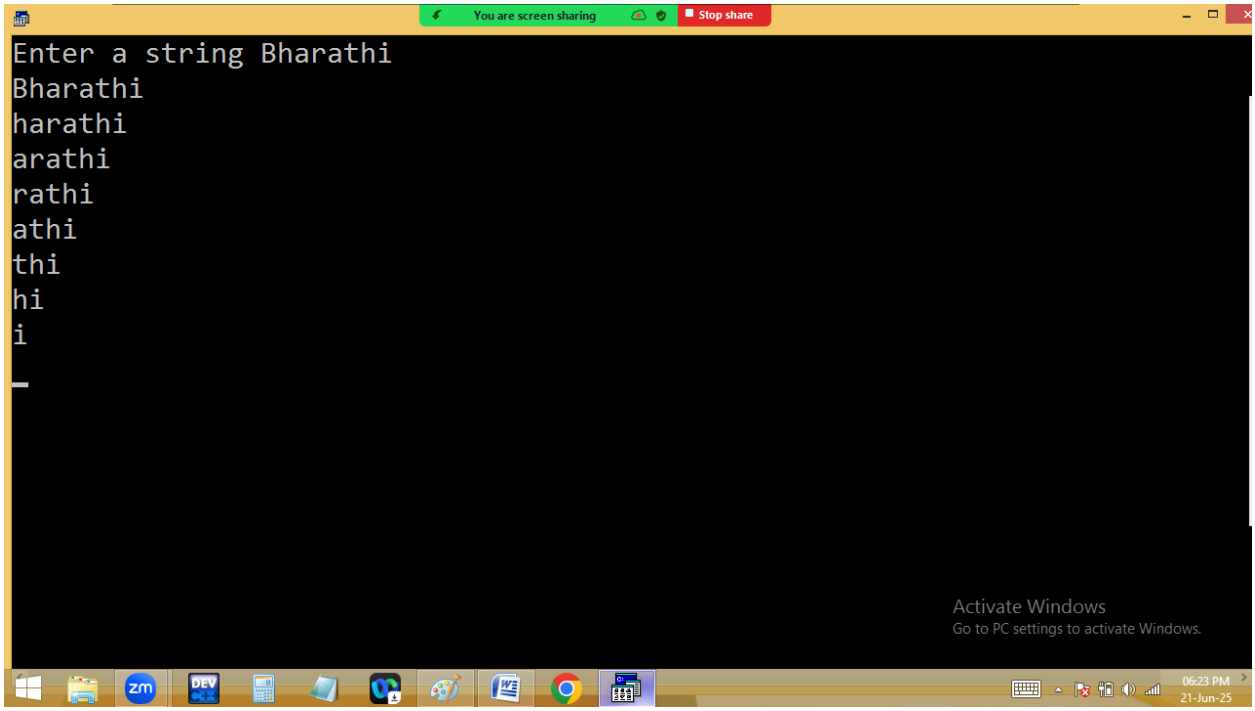
for(i=0; i<3; i++) p("%4d",a[i]);

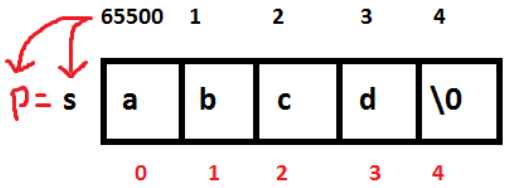




Pointer to string:



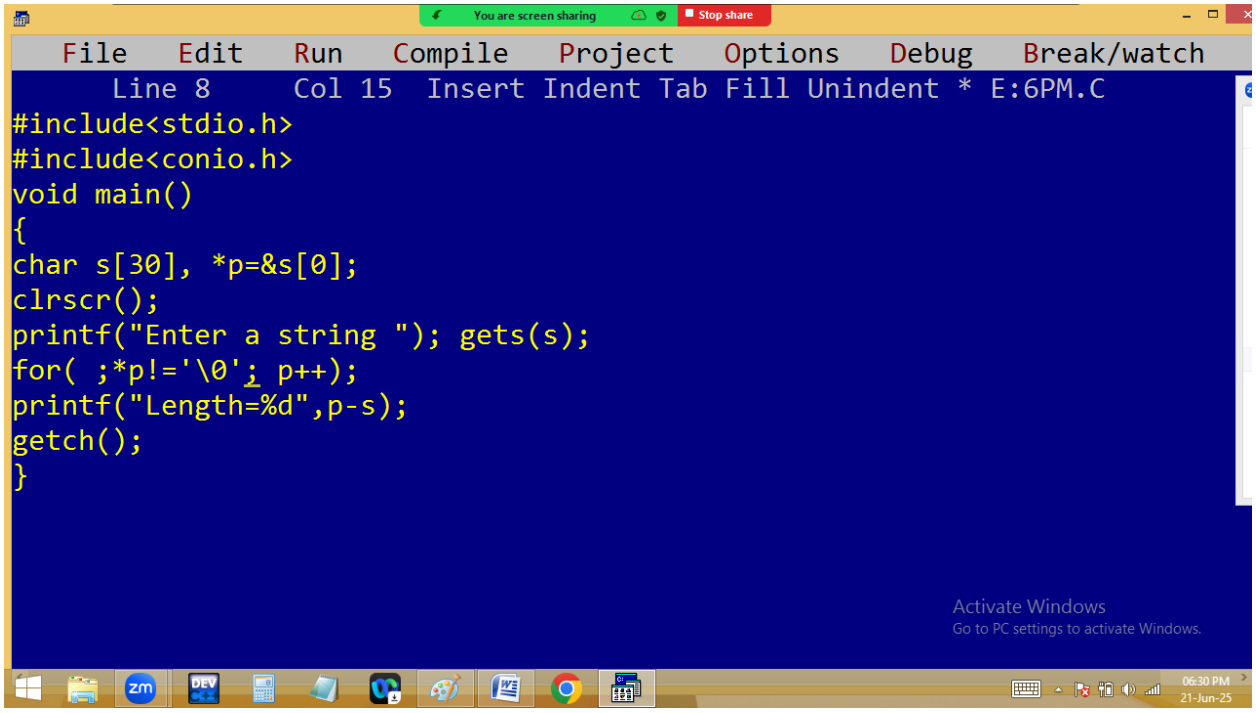


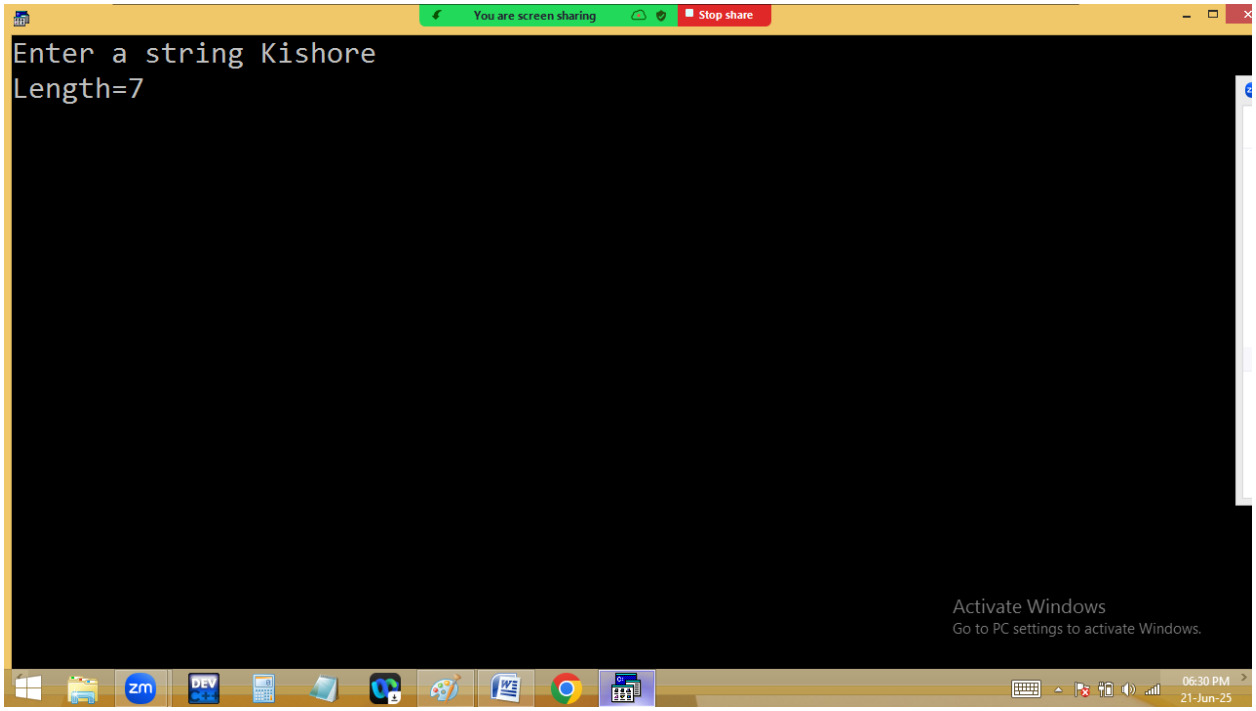


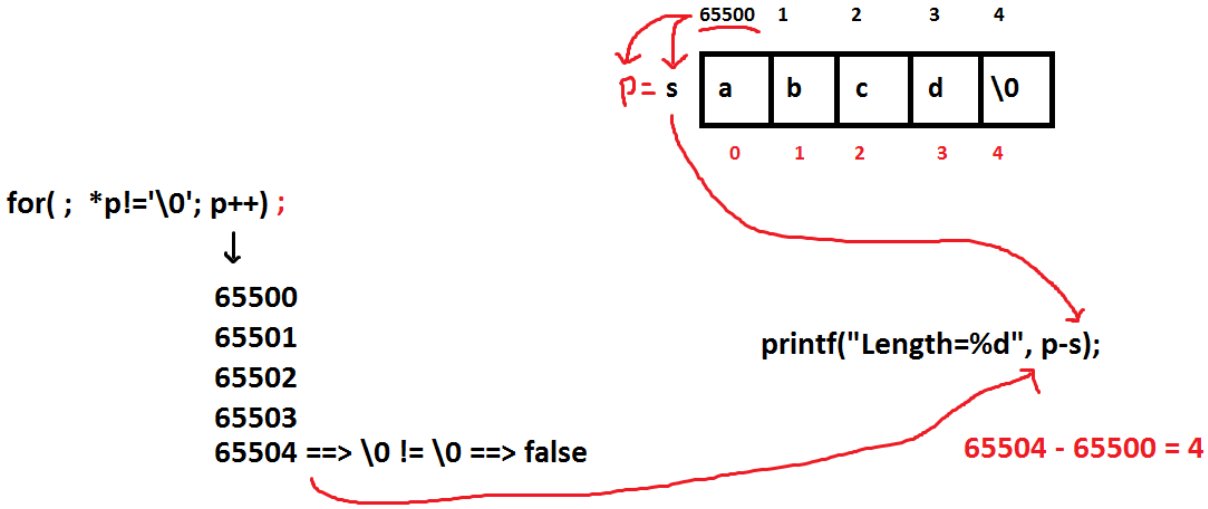

```
for( ;  *p!='\0'; p++) puts(p);
```

print 65500 to \0 ==> abcd
print 65501 to \0 ==> bcd
print 65502 to \0 ==> cd
print 65503 to \0 ==> d
65504 ==> \0 != \0 ==> false

Find the string length using pointer only?

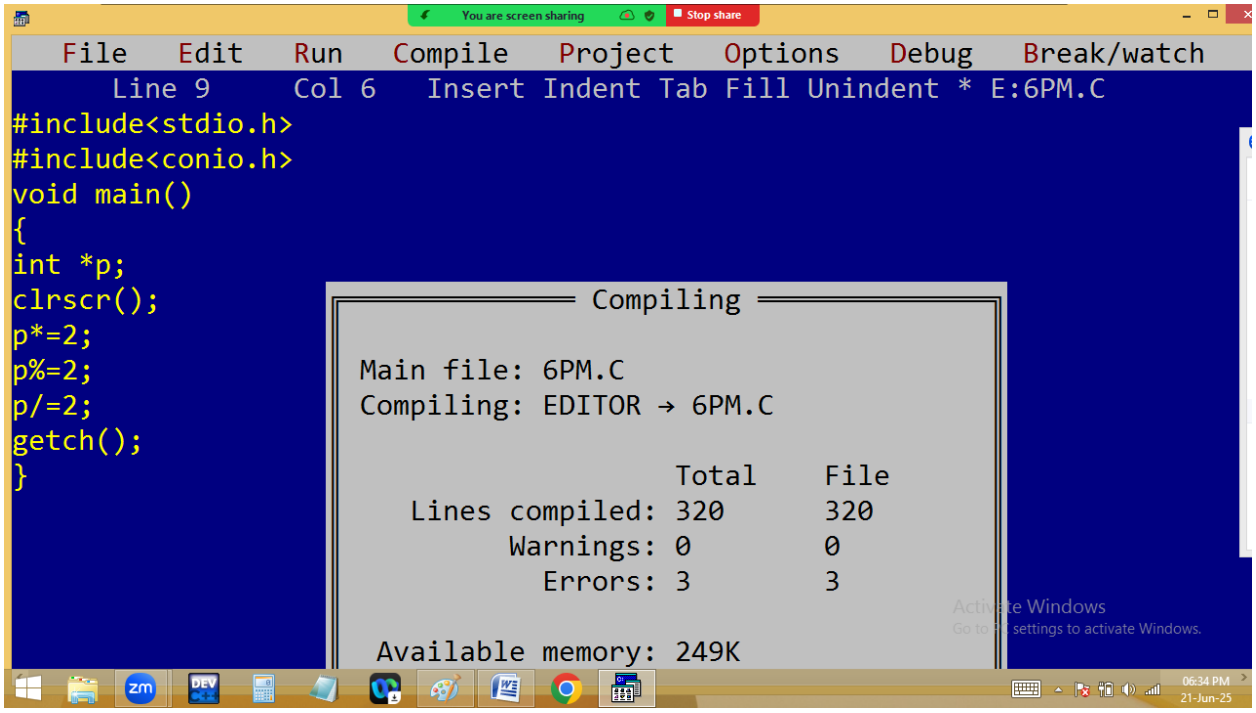

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File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 15 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char s[30], *p=&s[0];
clrscr();
printf("Enter a string "); gets(s);
for(;*p!='\0'; p++);
printf("Length=%d",p-s);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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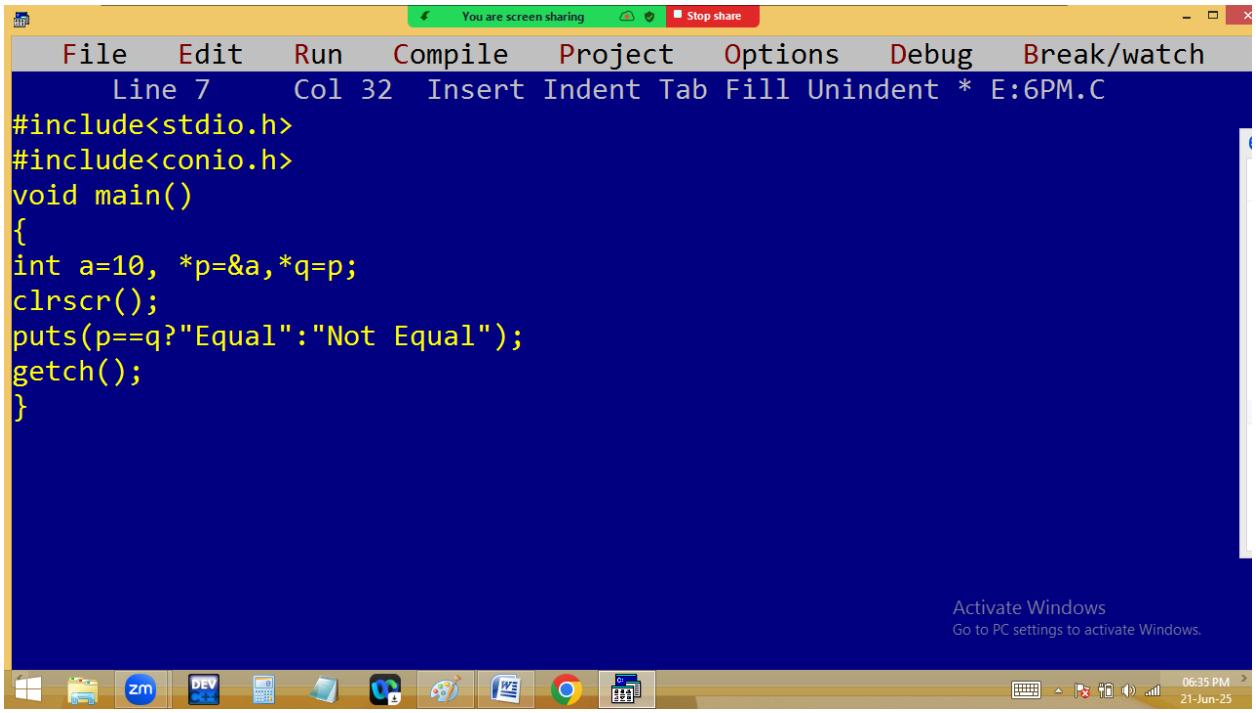


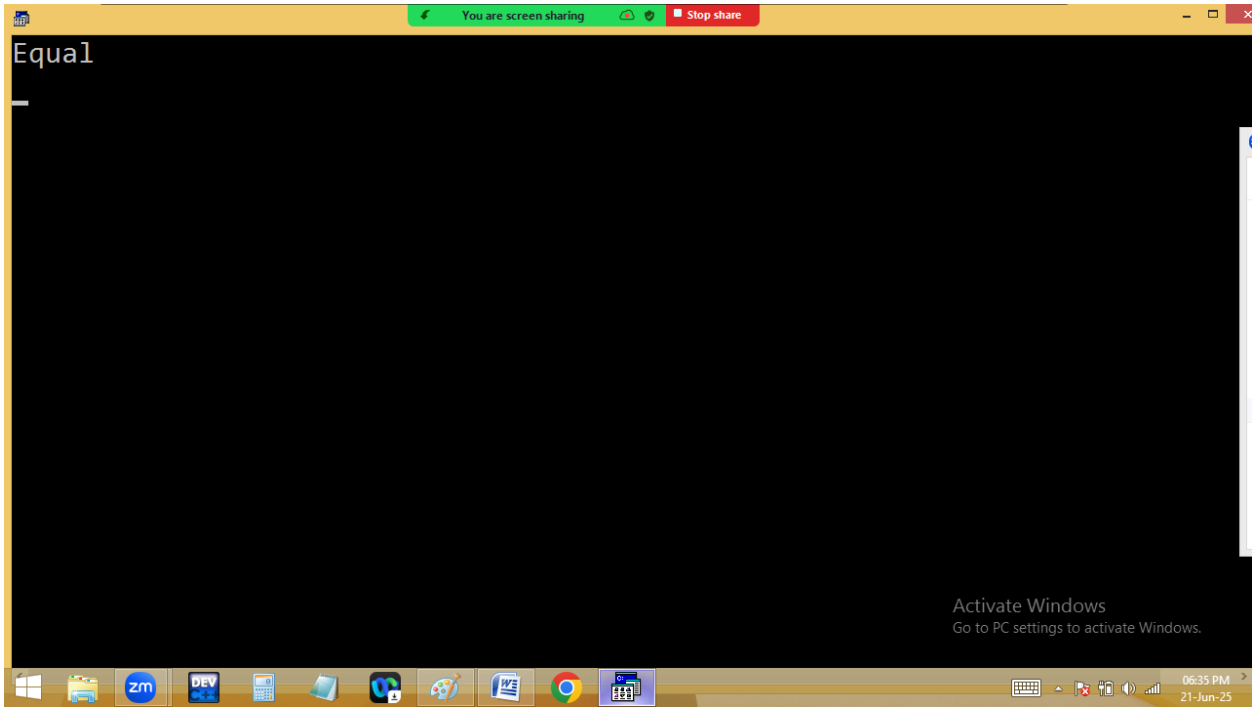


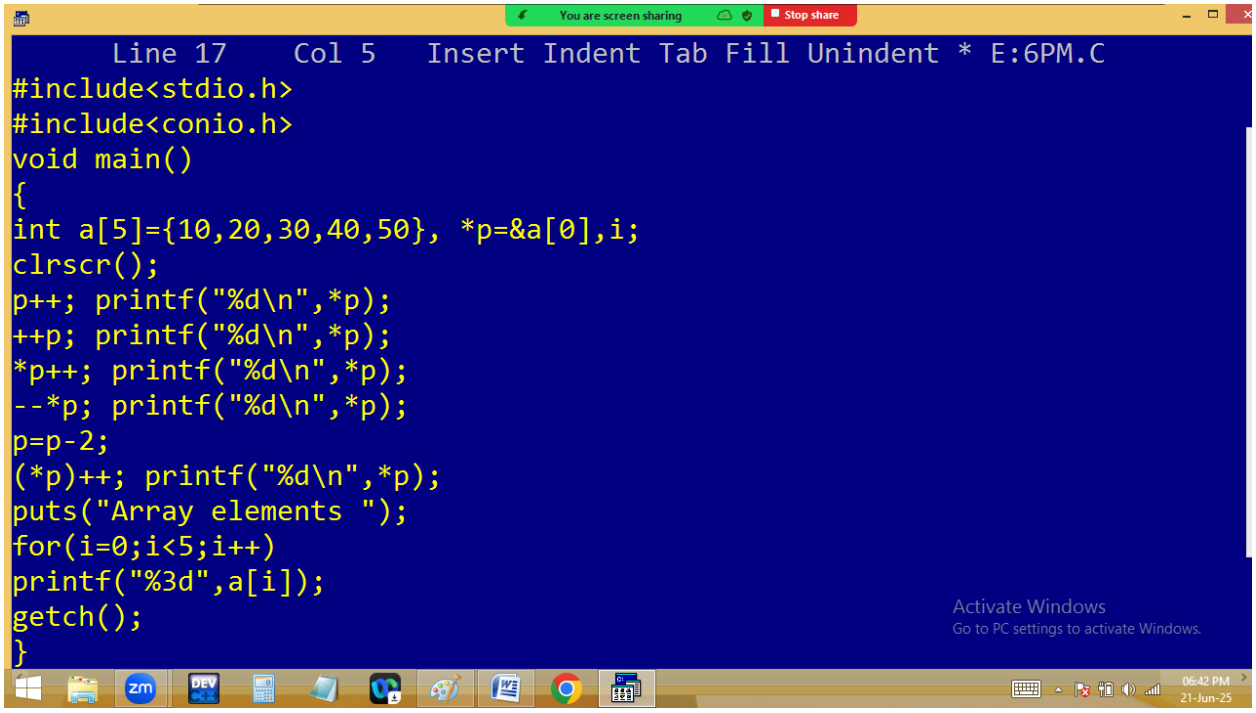
Pointer arithmetic:

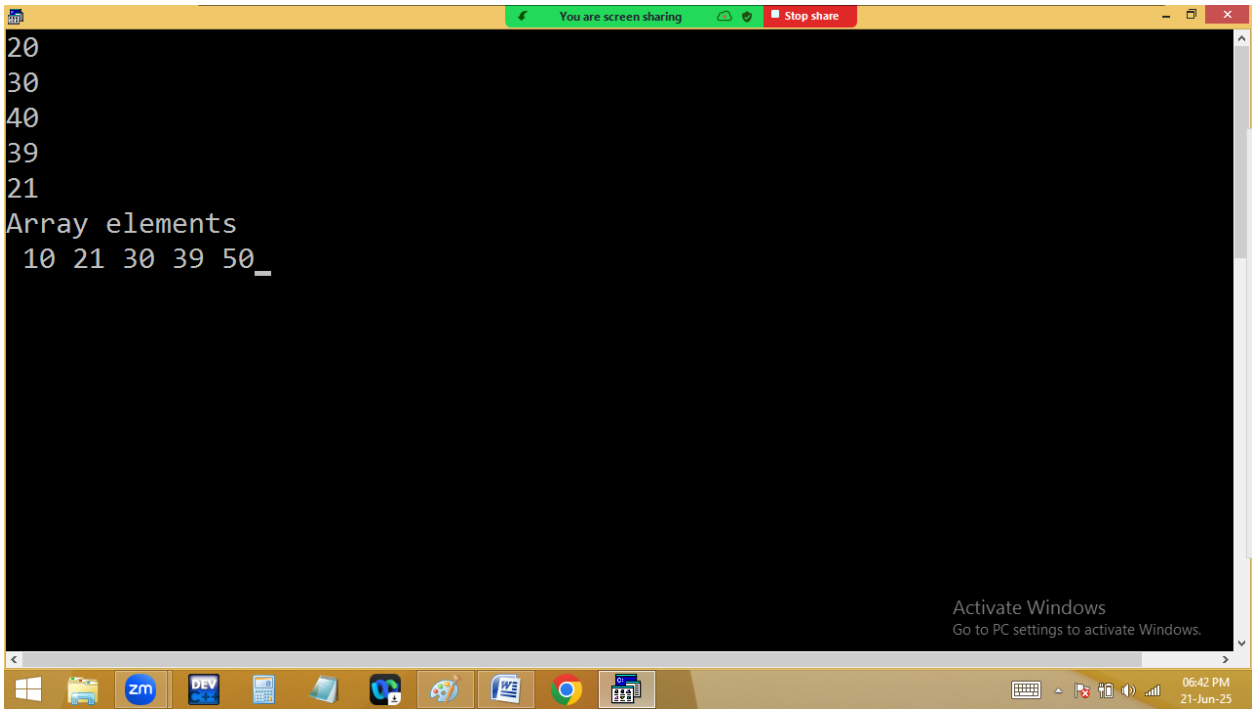
Like normal variables we can do =, ==, +, -, , ++, -- on pointers also. But we can't do *, %, / on pointers.


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File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 6 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int *p;
clrscr();
p*=2;
p%=2;
p/=2;
getch();
}
Compiling
Main file: 6PM.C
Compiling: EDITOR → 6PM.C
Total File
Lines compiled: 320 320
Warnings: 0 0
Errors: 3 3
Available memory: 249K
Activate Windows
06:34 PM
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File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 32 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10, *p=&a,*q=p;
clrscr();
puts(p==q?"Equal":"Not Equal");
getch();
}
Activate Windows
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You are screen sharing
Stop share
Equal
Activate Windows
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You are screen sharing
Stop share
Line 17 Col 5 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p=&a[0],i;
clrscr();
p++; printf("%d\n",*p);
++p; printf("%d\n",*p);
*p++; printf("%d\n",*p);
--*p; printf("%d\n",*p);
p=p-2;
(*p)++; printf("%d\n",*p);
puts("Array elements ");
for(i=0;i<5;i++)
printf("%3d",a[i]);
getch();
}
Activate Windows
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06:42 PM
21-Jun-25


You are screen sharing
Stop share
20
30
40
39
21
Array elements
10 21 30 39 50
Activate Windows
Go to PC settings to activate Windows.
06:42 PM
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p=65494

p++; 65494+1*2=65496

p(*p)==>value at 65496==> 20

++p==>65496+1*2=65498

p(*p)==>value at 65498==> 30

*p++ ==> p++ ==>65498+1*2=65500

p(*p)==> value at 65500 ==> 40

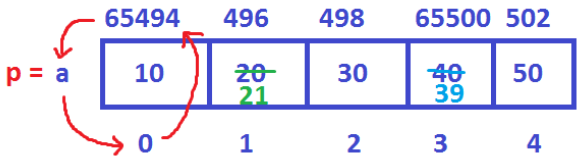
--*p==> -- value at 65500 ==> --40 ==> 39

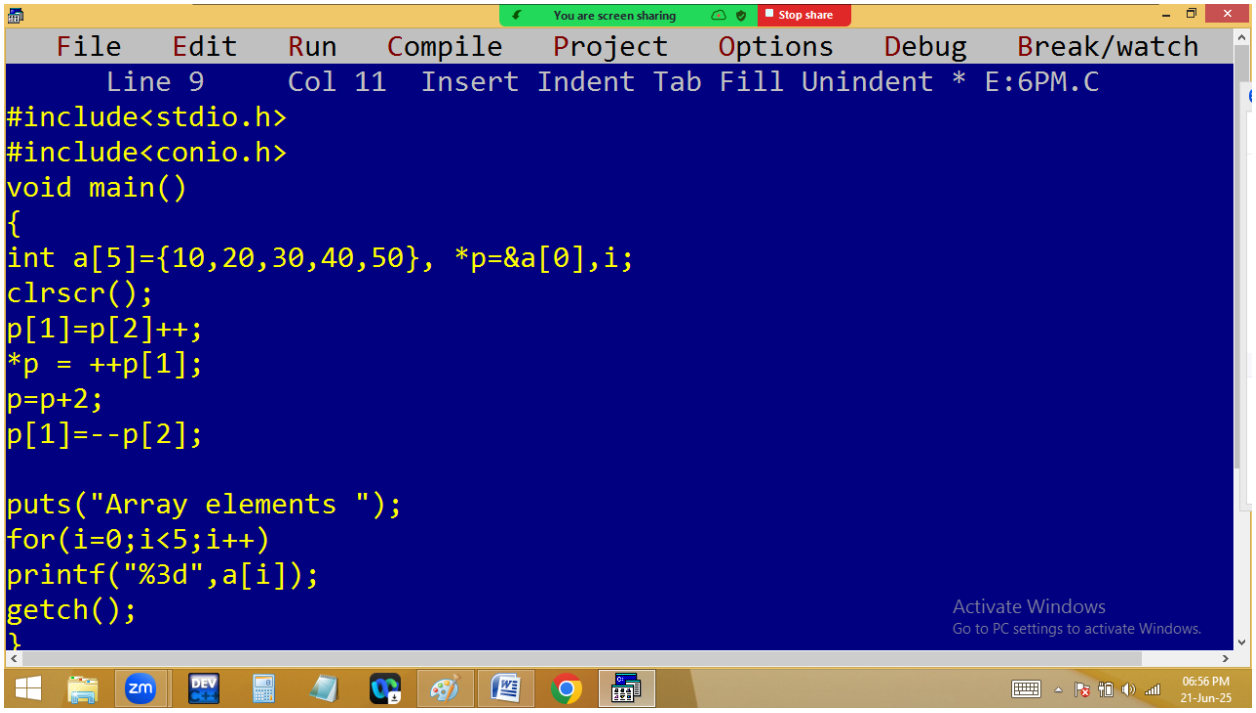
p(*p)==>value at 65500 ==> 39

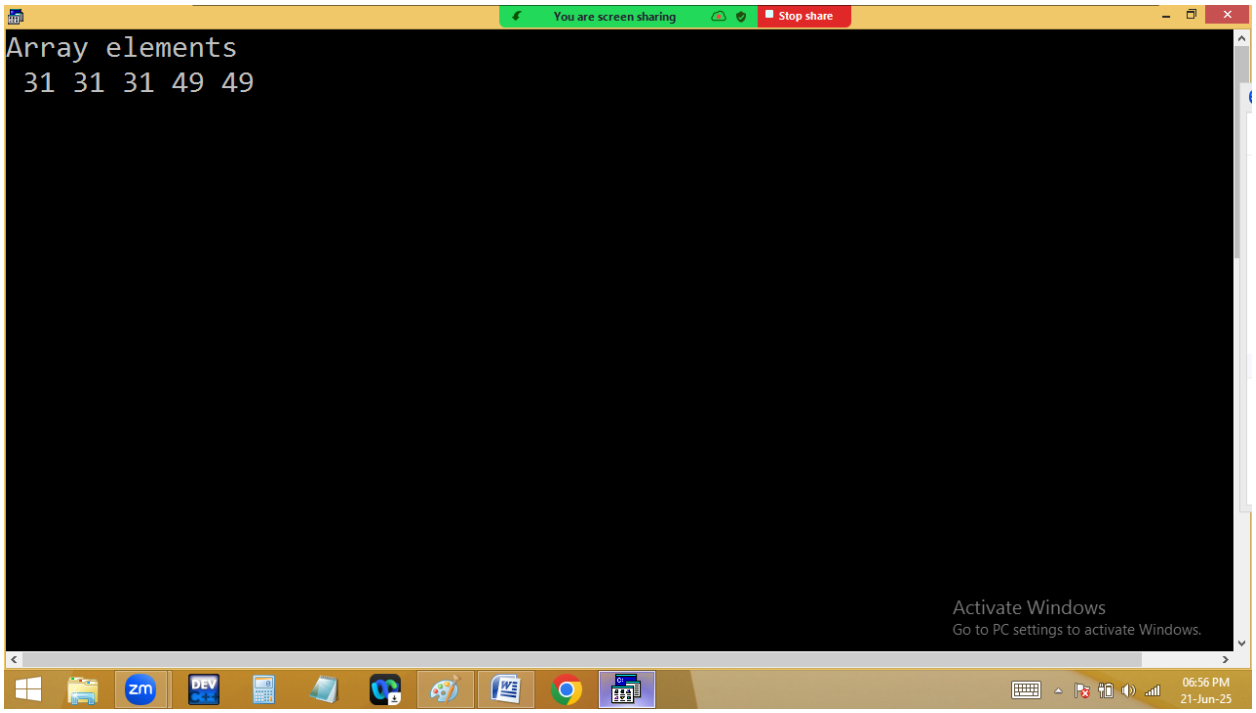
p=p-2==>65500-2*2=65496

(*p)++==>value at 65496++ ==> 20++ ==> 21

p(*p)==> value at 65496 ==> 21




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File Edit Run Compile Project Options Debug Break/watch
Line 9 Col 11 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p=&a[0],i;
clrscr();
p[1]=p[2]++;
*p = ++p[1];
p=p+2;
p[1]=--p[2];
puts("Array elements ");
for(i=0;i<5;i++)
printf("%3d",a[i]);
getch();
}
Activate Windows
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06:56 PM
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You are screen sharing
Stop share
Array elements
31 31 31 49 49
Activate Windows
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06:56 PM
21-Jun-25

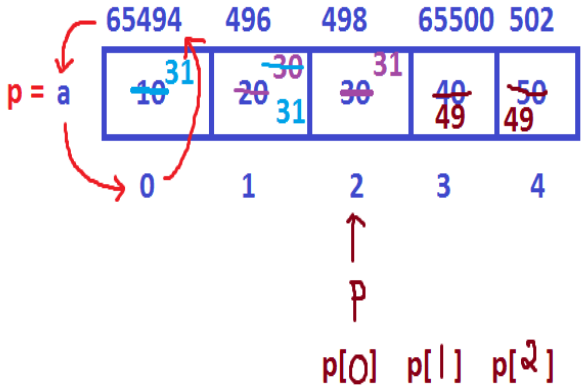
p=65494

p[1]=p[2]++; 1. p[1]=p[2] ==> p[1]=30

2. p[2]++ ==> 30++ ==> 31

*p = ++p[1]; 1. ++p[1] ==> 30++ ==> 31

2. *p = 31 ==> 65494 ==> p[0]= 31



p=p+2==>65495+2*2=65498

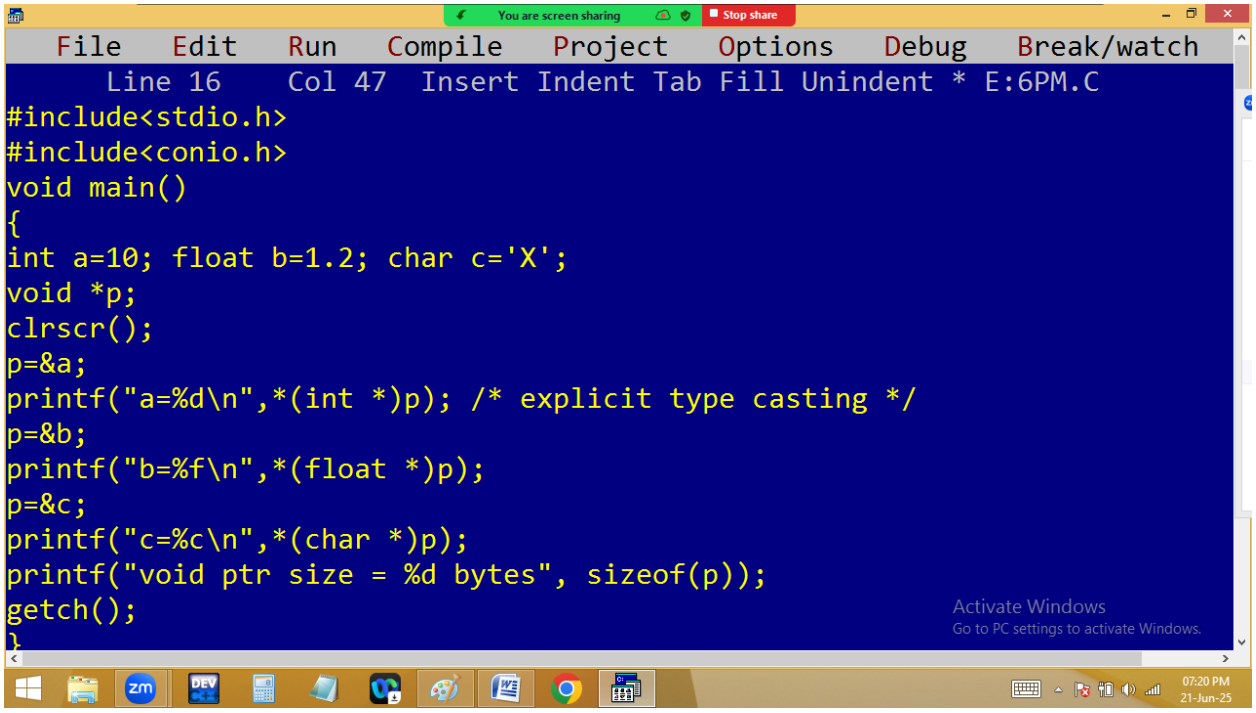
p=65498

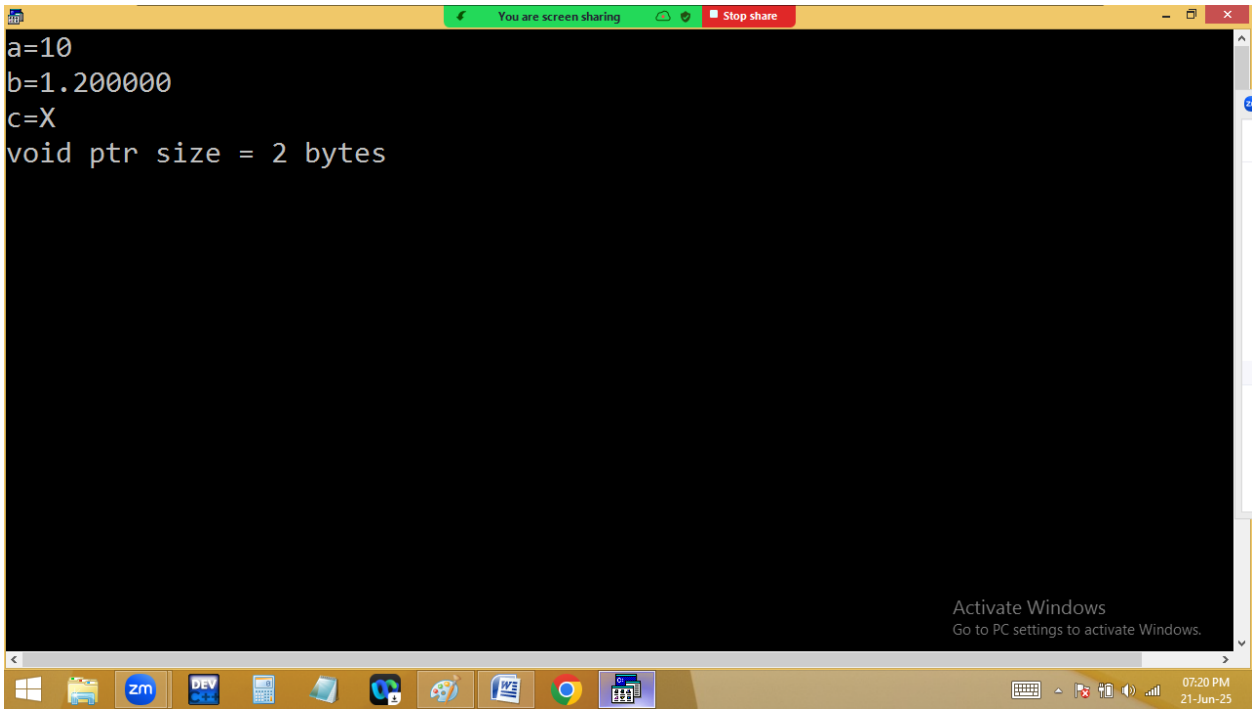
p[1]=--p[2]; 1. --p[2] ==> 65498+2*2=65502-- ==> 50 ==> 50-- ==> 49

2. p[1] = p[2] ==> 65500 = 65502 ==> 40 = 49

Void / generic pointer:

Pointer can store only the same type of variable address. Using void pointer we can store any type of variable address. But before going to access void pointer explicit type casting is compulsory. Void pointer takes 2 bytes. It is used to handle dynamic arrays.


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File Edit Run Compile Project Options Debug Break/watch
Line 16 Col 47 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; float b=1.2; char c='X';
void *p;
clrscr();
p=&a;
printf("a=%d\n",*(int *)p); /* explicit type casting */
p=&b;
printf("b=%f\n",*(float *)p);
p=&c;
printf("c=%c\n",*(char *)p);
printf("void ptr size = %d bytes", sizeof(p));
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
21-Jun-25


You are screen sharing
Stop share
a=10
b=1.200000
c=X
void ptr size = 2 bytes
Activate Windows
Go to PC settings to activate Windows.
07:20 PM
21-Jun-25